

Structure Of Atom

Question1

Energy and radius of first Bohr orbit of He^+ and Li^{2+} are [Given $R_H = 2.18 \times 10^{-18} \text{ J}$, $a_0 = 52.9 \text{ pm}$]

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Options:

A. $E_n(\text{Li}^{2+}) = -19.62 \times 10^{-16} \text{ J};$
 $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$
 $E_n(\text{He}^+) = -8.72 \times 10^{-16} \text{ J};$
 $r_n(\text{He}^+) = 26.4 \text{ pm}$

B. $E_n(\text{Li}^{2+}) = -8.72 \times 10^{-16} \text{ J};$
 $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$
 $E_n(\text{He}^+) = -19.62 \times 10^{-16} \text{ J};$
 $r_n(\text{He}^+) = 17.6 \text{ pm}$

C. $E_n(\text{Li}^{2+}) = -19.62 \times 10^{-18} \text{ J};$
 $r_n(\text{Li}^{2+}) = 17.6 \text{ pm}$
 $E_n(\text{He}^+) = -8.72 \times 10^{-18} \text{ J};$
 $r_n(\text{He}^+) = 26.4 \text{ pm}$

D. $E_n(\text{Li}^{2+}) = -8.72 \times 10^{-18} \text{ J};$
 $r_n(\text{Li}^{2+}) = 26.4 \text{ pm}$
 $E_n(\text{He}^+) = -19.62 \times 10^{-18} \text{ J};$
 $r_n(\text{He}^+) = 17.6 \text{ pm}$

Answer: C

Solution:

The energy and radius of the first Bohr orbit for a hydrogen-like atom is given by:

Energy:

$$E_n = \frac{-2.18 \times 10^{-18} \times Z^2}{n^2} \text{ J}$$

Radius:

$$r_n = \frac{52.9 \times n^2}{Z} \text{ pm}$$

Where:

Z is the atomic number.

n is the orbit number (or principal quantum number).

Let's calculate these values for the ions He^+ and Li^{2+} :

For He^+ :

The atomic number, $Z = 2$

Principal quantum number, $n = 1$

Energy, E_{He^+} :

$$E_{\text{He}^+} = -2.18 \times 10^{-18} \times 2^2 = -8.72 \times 10^{-18} \text{ J}$$

Radius, r_{He^+} :

$$r_{\text{He}^+} = \frac{52.9 \times 1^2}{2} = 26.45 \text{ pm}$$

For Li^{2+} :

The atomic number, $Z = 3$

Principal quantum number, $n = 1$

Energy, $E_{\text{Li}^{2+}}$:

$$E_{\text{Li}^{2+}} = -2.18 \times 10^{-18} \times 3^2 = -19.62 \times 10^{-18} \text{ J}$$

Radius, $r_{\text{Li}^{2+}}$:

$$r_{\text{Li}^{2+}} = \frac{52.9 \times 1^2}{3} = 17.63 \text{ pm}$$

These calculations show the energy and radius for the first Bohr orbit of He^+ and Li^{2+} .

Question2

The ratio of the wavelengths of the light absorbed by a Hydrogen atom when it undergoes $n = 2 \rightarrow n = 3$ and $n = 4 \rightarrow n = 6$ transitions, respectively, is

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Options:

A. $\frac{1}{9}$

B. $\frac{1}{4}$

C. $\frac{1}{36}$

D. $\frac{1}{16}$

Answer: B

Solution:

To find the ratio of the wavelengths of the light absorbed by a hydrogen atom during the transitions $n = 2 \rightarrow n = 3$ and $n = 4 \rightarrow n = 6$, we start by calculating the change in energy (ΔE) for each transition.

The energy of an electron in a hydrogen atom at a given level n is given by:

$$E_n = \frac{-R_H}{n^2}$$

where R_H is the Rydberg constant.

Transition $n = 2 \rightarrow n = 3$:

$$\Delta E_{2 \rightarrow 3} = E_3 - E_2 = \frac{-R_H}{3^2} - \left(\frac{-R_H}{2^2} \right)$$

$$= R_H \left(\frac{1}{4} - \frac{1}{9} \right)$$

$$= R_H \times \frac{5}{36}$$

The wavelength $\lambda_{2 \rightarrow 3}$ is then:

$$\lambda_{2 \rightarrow 3} = \frac{hc}{\Delta E_{2 \rightarrow 3}} = \frac{hc \cdot 36}{R_H \cdot 5}$$

Transition $n = 4 \rightarrow n = 6$:

$$\Delta E_{4 \rightarrow 6} = E_6 - E_4 = \frac{-R_H}{6^2} + \frac{R_H}{4^2}$$

$$= \frac{R_H \times 20}{36 \times 16}$$

The wavelength $\lambda_{4 \rightarrow 6}$ is then:

$$\lambda_{4 \rightarrow 6} = \frac{hc}{\Delta E_{4 \rightarrow 6}} = \frac{hc \times 36 \times 16}{R_H \cdot 20}$$

Calculating the ratio:

The ratio of the wavelengths $\frac{\lambda_{2 \rightarrow 3}}{\lambda_{4 \rightarrow 6}}$ is:

$$\begin{aligned} \frac{\lambda_{2 \rightarrow 3}}{\lambda_{4 \rightarrow 6}} &= \frac{\frac{hc \cdot 36}{R_H \cdot 5}}{\frac{hc \times 36 \times 16}{R_H \cdot 20}} \\ &= \frac{1}{4} \end{aligned}$$

Therefore, the ratio of the wavelengths for the given transitions is $\frac{1}{4}$.

Question3

The quantum numbers of four electrons are given below :

I. $n = 4; l = 2; m_l = -2; s = -\frac{1}{2}$

II. $n = 3; l = 2; m_l = 1; s = +\frac{1}{2}$

III. $n = 4; l = 1; m_l = 0; s = +\frac{1}{2}$

IV. $n = 3; l = 1; m_l = -1; s = +\frac{1}{2}$

The correct decreasing order of energy of these electrons is

[NEET 2024 Re]

Options:

A.

$$IV > II > III > I$$

B.

$$I > III > II > IV$$

C.

$$III > I > II > IV$$

D.

$$I > II > III > IV$$

Answer: B

Solution:

(I) $n = 4, l = 2, m_l = -2, s = -\frac{1}{2}$; represents $4d(n + l = 6)$

(II) $n = 3, l = 2, m_l = 1, s = +\frac{1}{2}$; represents $3d(n + l = 5)$

(III) $n = 4, l = 1, m_l = 0, s = +\frac{1}{2}$; represents $4p(n + l = 5)$

(IV) $n = 3, l = 1, m_l = -1, s = +\frac{1}{2}$; represents $3p(n + l = 4)$

Order of energy depends on the $(n + l)$, greater is the $(n + l)$ value greater is the energy, if $(n + l)$ is same, then it depends on n ; if 'n' is more, energy is more.

Step-1: According to $(n + l)$

$$\text{Energy} = (I) > (II) = (III) > (IV)$$

Step-2 : If $n \uparrow$, then energy increases

$$\text{Energy} = (I) > (III) > (II) > (IV)$$

Question4

Given below are two statements:

Statement I: The Balmer spectral line for H atom with lowest energy is located at $\frac{5}{36}R_H \text{ cm}^{-1}$.

(R_H = Rydberg constant)

Statement II: When the temperature of blackbody increases, the maxima of the curve (intensity and wavelength) shifts to shorter wavelength.

In the light of the above statements, choose the correct answer from the options given below:

[NEET 2024 Re]

Options:

A.

Statement I is true but Statement II is false

B.

Statement I is false but Statement II is true

C.

Both Statement I and Statement II are true

D.

Both Statement I and Statement II are false

Answer: C

Solution:

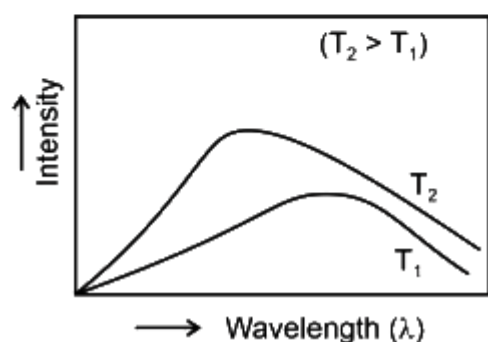
For Balmer series $n_1 = 2$

$$n_2 = 3$$

$$\bar{\nu} = \frac{1}{\lambda} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\bar{\nu} = R_H \left[\frac{1}{4} - \frac{1}{9} \right]$$

$$\bar{\nu} = \frac{5R_H}{36} \text{ cm}^{-1}$$



Question5

The energy of an electron in the ground state ($n = 1$) for He^+ ion is $-x\text{J}$, then that for an electron in $n = 2$ state for Be^{3+} ion in J is

[NEET 2024]

Options:

A.

$$-x$$

B.

$$-\frac{x}{9}$$

C.

$$-4x$$

D.

$$-\frac{4}{9}x$$

Answer: A

Solution:

$$E_n = -R_H \left(\frac{Z^2}{n^2} \right) J$$

For He* ($n = 1$),

$$E_n = -x = -R_H \left(\frac{2^2}{1^2} \right) = -4R_H$$

$$\therefore R_H = \frac{x}{4}$$

For Be³⁺ ($n = 2$),

$$E_n = -R_H \left(\frac{Z^2}{n^2} \right) J$$

$$= -\frac{x}{4} \times \left(\frac{4 \times 4}{2 \times 2} \right) = -xJ$$

Question6

Match List I with List II

	List I (Quantum Number)		List II (Information provided)
A.	m _l	I.	Shape of orbital
B.	m _s	II.	Size of orbital
C.	l	III.	Orientation of orbital
D.	n	IV.	Orientation of spin of electron

Choose the correct answer from the options given below :

[NEET 2024]

Options:

A.

A-I, B-III, C-II, D-IV

B.

A-III, B-IV, C-I, D-II

C.

A-III, B-IV, C-II, D-I

D.

A-II, B-I, C-IV, D-III

Answer: B

Solution:

- Magnetic quantum number m_l informs about orientation of orbital.
 - Spin quantum number m_s informs about orientation of spin of electron.
 - Azimuthal quantum number (l) informs about shape of orbital
 - Principal quantum number (n) informs about size of orbital
-

Question7

Select the correct statements from the following

A. Atoms of all elements are composed of two fundamental particles.

B. The mass of the electron is 9.10939×10^{-31} kg.

C. All the isotopes of a given element show same chemical properties:

D. Protons and electrons are collectively known as nucleons.

E. Dalton's atomic theory, regarded the atom as an ultimate particles of matter

Choose the correct answer from the options given below

[NEET 2023]

Options:

A.

C, D and E only

B.

A and E only

C.

B, C and E only

D.

A, B and C only

Answer: C

Solution:

- Atoms consist of three fundamental particles :

Electrons, protons and neutrons

- The mass of the electron is $9.10939 \times 10^{-31} \text{ kg}$
- All the isotopes of a given element show same chemical properties.
- Protons and neutrons present in the nucleus are collectively called as nucleons.
- Dalton's atomic theory, regarded the atom as the ultimate particle of matter

So, the correct statements are B, C, E only

Question8

The relation between n_m , ($n_m =$. the number of permissible values of magnetic quantum number (m)) for a given value of azimuthal quantum number (l), is

[NEET 2023]

Options:

A.

$$l = 2n_m + 1$$

B.

$$n_m = 2l + 1$$

C.

$$n_m = l + 2$$

D.

$$I = \frac{n_m - 1}{2}$$

Answer: D

Solution:

Values of n_m (magnetic quantum number) for given azimuthal quantum number can be calculated as following

$$n_m = 2I + 1$$

$$I = \frac{n_m - 1}{2}$$

Question9

Incorrect set of quantum numbers from the following is :

[NEET 2023 mpr]

Options:

A.

$$n = 4, l = 3, m_l = -3, -2, -1, 0, +1, +2, +3,$$

$$m_s = -1/2$$

B.

$$n = 5, l = 2, m_l = -2, -1, +1, +2, m_s = +1/2$$

C.

$$n = 4, l = 2, m_l = -2, -1, 0, +1, +2,$$

$$m_s = -1/2$$

D.

$$n = 5, l = 3, m_l = -3, -2, -1, 0, +1, +2, +3$$

$$m_s = +1/2$$

Answer: B

Solution:

$$n = 5, l = 2, m = -2, -1, +1, +2, m_s = +\frac{1}{2}$$

Question10

Given below are two statements :

Statement I : The value of wave function, Ψ depends upon the coordinates of the electron in the atom.

Statement II : The probability of finding an electron at a point within an atom is proportional to the orbital wave function.

In the light of the above statements, choose the correct answer from the options given below :

[NEET 2023 mpr]

Options:

A.

Statement I is true but Statement II is false.

B.

Statement I is false but Statement II is true.

C.

Both Statement I and Statement II are true.

D.

Both Statement I and Statement II are false.

Answer: A

Solution:

Statement-I is true and statement-II is false.

Question11

Match List - I with List - II :

List - I (quantum number)	List - II (orbital)
(a) $n = 2, l = 1$	(i) 2 s
(b) $n = 3, l = 2$	(ii) 3 s
(c) $n = 3, l = 0$	(iii) 2 p
(d) $n = 2, l = 0$	(iv) 3 d

**Choose the correct answer from the options given below :
[NEET Re-2022]**

Options:

- A. (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
- B. (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)
- C. (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)
- D. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

Answer: A

Solution:

Solution:

$\ell = 0 \Rightarrow$ s-subshell

$\ell = 1 \Rightarrow$ p-subshell

$\ell = 2 \Rightarrow$ d-subshell

$\ell = 3 \Rightarrow$ f-subshell

$\therefore n = 2, \ell = 1 \Rightarrow 2p$

$n = 3, \ell = 2 \Rightarrow 3d$

$n = 3, \ell = 0 \Rightarrow 3s$

$n = 2, \ell = 0 \Rightarrow 2s$

Question12

When electromagnetic radiation of wavelength 300 nm falls on the surface of a metal, electrons are emitted with the kinetic energy of $1.68 \times 10^5 \text{ J mol}^{-1}$. What is the minimum energy needed to remove an electron from the metal ?

($h = 6.626 \times 10^{-34} \text{ Js}$, $c = 3 \times 10^8 \text{ ms}^{-1}$.,

$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$)

[NEET Re-2022]

Options:

A. $2.31 \times 10^5 \text{ J l}^{-1}$

B. $2.31 \times 10^6 \text{ J l}^{-1}$

C. $3.84 \times 10^4 \text{ J l}^{-1}$

D. $3.84 \times 10^{-19} \text{ J l}^{-1}$

Answer: A

Solution:

Solution:

$$E_P = \phi + K \cdot E$$

$$E_P - K \cdot E = \phi$$

$$E_P = \frac{hc}{\lambda} = 3.975 \times 10^5 \text{ Jmol}^{-1}$$

$$K \cdot E = 1.68 \times 10^5 \text{ Jmol}^{-1}$$

$$\begin{aligned}\phi &= (3.975 - 1.68) \times 10^5 \\ &= 2.295 \times 10^5 \approx 2.31 \times 10^5 \text{ Jmol}^{-1}\end{aligned}$$

Question13

Identify the incorrect statement from the following.
[NEET-2022]

Options:

- A. All the five 5d orbitals are different in size when compared to the respective 4d orbitals.
- B. All the five 4d orbitals have shapes similar to the respective 3d orbitals.
- C. In an atom, all the five 3d orbitals are equal in energy in free state.
- D. The shapes of d_{xy} , d_{yz} and d_{zx} orbitals are similar to each other; and $d_{x^2-y^2}$ and d_{z^2} are similar to each other.

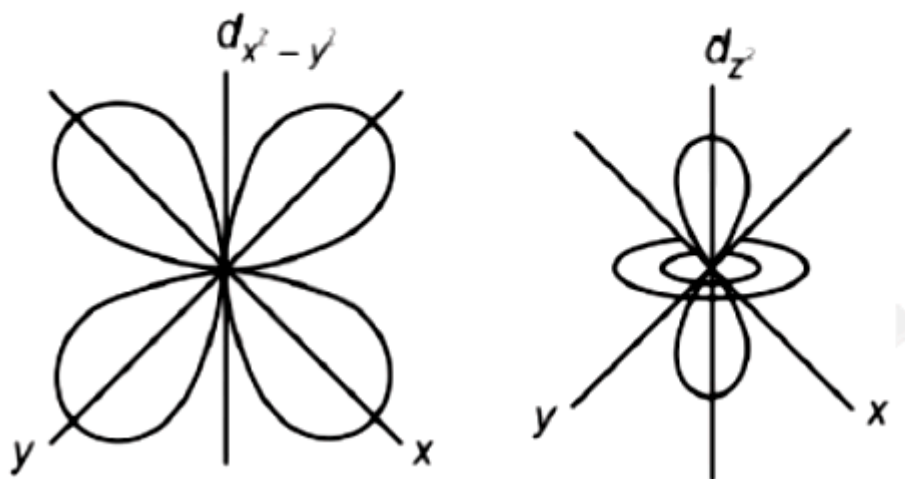
Answer: D

Solution:

Solution:

In an atom, all the five $3d$ orbitals are equal in energy in free state i.e., degenerate.

- The shape of $d_{x^2-y^2}$ is different than shape of d_{z^2}



- The size of orbital depends on principal quantum number 'n' therefore all the five $3d$ orbitals are different in size when compared to the respective $4d$ orbitals.

- Shape of orbitals depends on azimuthal quantum number 'l' therefore shapes of $4d$ orbitals are similar to the respective $3d$ orbitals.

Question14

If radius of second Bohr orbit of the He^+ ion is 105.8pm , what is the radius of third Bohr orbit of Li^{2+} ion?
[NEET-2022]

Options:

- A. 158.7pm
- B. 15.87pm
- C. 1.587pm
- D. $158.7\text{\AA}$

Answer: A

Solution:

Solution:

$$r_n \propto \frac{n^2}{Z}$$

$$\frac{r_3(Li^{2-})}{r_2(He^-)} = \frac{(n_3)^2}{Z(Li^{2-})} \times \frac{Z(He^+)}{(n_2)^2}$$

$$\frac{r_3(Li^{2+})}{105.8} = \frac{(3)^2}{3} \times \frac{2}{(2)^2}$$

$$= 105.8 \times \frac{3}{2}$$

$$r_3(Li^{2+}) = 158.7 pm$$

Question15

A particular station of All India Radio, New Delhi broadcasts on a frequency of 1, 368 kH z (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is : [speed of light $c = 3.0 \times 10^8 \text{ ms}^{-1}$]
[NEET 2021]

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Options:

- A. 219.3 m
- B. 219.2 m
- C. 2192 m
- D. 21.92 cm

Answer: A

Solution:

Solution:

Energy of electromagnetic radiation (E)

$$= \frac{hc}{\lambda} = h\nu$$

$$\text{So, } \frac{c}{\lambda} = \gamma \Rightarrow \lambda = \frac{c}{\gamma}$$

$$\lambda = \frac{3 \times 10^8}{1368 \times 10^3} = 219.3 \text{ m}$$

Question16

4d , 5p, 5f and 6p orbitals are arranged in the order of decreasing energy. The correct option is (NEET 2019)

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Options:

A. $5f > 6p > 4d > 5p$

B. $5f > 6p > 5p > 4d$

C. $6p > 5f > 5p > 4d$

D. $6p > 5f > 4d > 5p$

Answer: B

Solution:

Solution:

Higher the value of $(n+1)$ for an orbital, higher is its energy. However, if two different types of orbitals have same value of $(n+1)$, the orbital with lower value of n has lower energy. Therefore, decreasing order of energy of the given orbitals is $5f > 6p > 5p > 4d$.

Question17

Which of the following series of transitions in the spectrum of hydrogen atom falls in visible region? (NEET 2019)

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Options:

- A. Brackett series
- B. Lyman series
- C. Balmer series
- D. Paschen series

Answer: C

Solution:

Solution:

Lyman series : UV region

Balmer series : Visible region

Paschen series : IR region

Brackett series : IR region

Question18

Orbital having 3 angular nodes and 3 total nodes is (Odisha NEET 2019)

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Options:

- A. 5p
- B. 3d
- C. 4f
- D. 6d

Answer: C

Solution:

Solution:

Number of spherical/radial nodes in any orbital $= n - 1 - 1$

Number of planar/angular nodes in orbital $= l = 3$

$\therefore$ Total number of nodes in any orbital $= n - 1 = 3$

$\therefore n = 4$

Thus, the orbital is 4f.

Question19

In hydrogen atom, the de Broglie wavelength of an electron in the second Bohr orbit is

[Given that Bohr radius, $a_0 = 52.9\text{pm}$]

(Odisha NEET 2019)

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Options:

A. 211.6pm

B. $211.6\pi\text{pm}$

C. $52.9\pi\text{pm}$

D. 105.8pm

Answer: B

Solution:

Solution:

Bohr radius, $a_0 = 52.9\text{pm}$

$n = 2$, $r_n = n^2 a_0 = (2)^2 a_0 = 4 \times 52.9\text{pm} = 211.6\text{pm}$

The angular momentum of an electron in a given stationary state can be expressed as in equation,

$$mvr = n \cdot \frac{h}{2\pi} = 2 \times \frac{h}{2\pi} = \frac{h}{\pi}$$

de-Broglie equation $mvr\pi = h \dots (i)$

$$\lambda = \frac{h}{mv}; \lambda mv = h \dots (ii)$$

From equations, (i) and (ii), we get $\lambda = \pi r$

Putting the value of r , $\lambda = 211.6\pi\text{pm}$

Question20

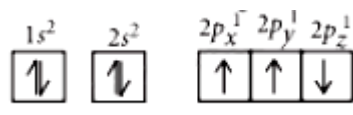
Which one is a wrong statement?

(NEET 2018)

Options:

- A. Total orbital angular momentum of electron in s -orbital is equal to zero.
- B. An orbital is designated by three quantum numbers while an electron in an atom is designated by four quantum numbers.
- C.

The electronic configuration of N atom is



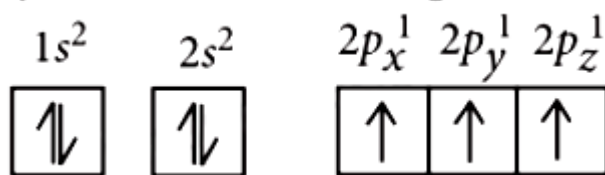
- D. The value of m for d_{z^2} is zero.

Answer: C

Solution:

Solution:

According to Hund's rule of maximum multiplicity, the correct configuration of 'N' is



Question21

**Which one is the wrong statement?
(NEET 2017)**

Options:

- A. (a) The uncertainty principle is $\Delta E \times \Delta t \geq \frac{h}{4\pi}$
- B. Half filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement.
- C. The energy of 2s -orbital is less than the energy of 2p -orbital in case of hydrogen like atoms.

D. de-Broglie's wavelength is given by $\lambda = \frac{h}{mv}$, where m = mass of the particle, v = group velocity of the particle.

Answer: C

Solution:

Solution:

In case of hydrogen like atoms, energy depends on the principal quantum number only. Hence, $2s$ - orbital will have energy equal to $2p$ -orbital.

Question22

**How many electrons can fit in the orbital for which $n = 3$ and $l = 1$?
(NEET- II 2016)**

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Options:

- A. 2
- B. 6
- C. 10
- D. 14

Answer: B

Solution:

Solution:

For $n = 3$ and $l = 1$, the subshell is $3p$ and a particular $3p$ orbital can accommodate 6 electrons.

Question23

**Which of the following pairs of d -orbitals will have electron density along the axes?
(NEET-II 2016)**

Options:

A. d_{z^2} , d_{xz}

B. d_{xz} , d_{yz}

C. d_{z^2} , $d_{x^2-y^2}$

D. d_{xy} , $d_{x^2-y^2}$

Answer: C

Solution:

Solution:

$d_{x^2-y^2}$ and d_{z^2} orbitals have electron density along the axes while d_{xy} , d_{yz} and d_{xz} orbitals have electron density in between the axes.

Question24

Two electrons occupying the same orbital are distinguished by (NEET -I 2016)

Options:

A. azimuthal quantum number

B. spin quantum number

C. principal quantum number

D. magnetic quantum number.

Answer: B

Solution:

Solution:

For the two electrons occupying the same orbital values of n , l and m_l are same but m_s is different, i.e., $+\frac{1}{2}$ and $-\frac{1}{2}$

Question25

Which is the correct order of increasing energy of the listed orbitals in the atom of titanium (At. no. $Z=22$) (2015)

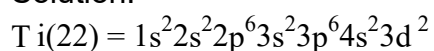
Options:

- A. $4s, 3s, 3p, 3d$
- B. $3s, 3p, 3d, 4s$
- C. $3s, 3p, 4s, 3d$
- D. $3s, 4s, 3p, 3d$

Answer: C

Solution:

Solution:



$\therefore$ Order of increasing energy is $3s, 3p, 4s, 3d$

Question26

The angular momentum of electron in 'd' orbitals is equal to (2015 Cancelled)

Options:

- A. $2\sqrt{3}\hbar$

B. $0\hbar$

C. $\sqrt{6}\hbar$

D. $\sqrt{2}\hbar$

Answer: C

Solution:

Solution:

Angular momentum = $\sqrt{l(l+1)}\hbar$

For d orbital, $l = 2$

Angular momentum = $\sqrt{2(2+1)}\hbar = \sqrt{6}\hbar$

Question27

**The number of d-electrons in Fe^{2+} ($Z = 26$) is not equal to the number of electrons in which one of the following ?
(2015 Cancelled)**

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Options:

A. d- electrons in Fe ($Z=26$)

B. p - electrons in Ne ($Z=10$)

C. s- electrons in Mg ($Z=12$)

D. p- electrons in Cl ($Z=17$)

Answer: D

Solution:

Solution:

Number of d-electrons in $\text{Fe}^{2+} = 6$

Number of p-electrons Cl=11

Question28

**Be^{2+} is isoelectronic with which of the following ions ?
(2014)**

Options:

A. H^+

B. Li^+

C. Na^+

D. Mg^{2+}

Answer: B

Solution:

Solution:

Species	No. of electrons
Be^{2+}	2
H^+	0
Li^+	2
Na^+	10
Mg^{2+}	10

Question29

**Calculate the energy in Joule corresponding to light of wave length 45nm (Planck's constant $h = 6.63 \times 10^{-34} \text{ J s}$, speed of light, $c = 3 \times 10^8 \text{ ms}^{-1}$)
(2014)**

Options:

A. $6.67 \times 10^{15} \text{ J}$

B. $6.67 \times 10^{11} \text{ J}$

C. $4.42 \times 10^{-15} \text{ J}$

D. $4.42 \times 10^{-18} \text{ J}$

Answer: D

Solution:

Solution:

$$E = \frac{hc}{\lambda} \quad [\text{Given } \lambda = 45\text{nm} = 45 \times 10^{-9}\text{m}]$$

On putting the given values in the equation, we get

$$E = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{45 \times 10^9} = 4.42 \times 10^{-18} \text{ J}$$

Question30

What is the maximum number of orbitals that can be identified with following quantum numbers ?

$$n = 3, l = 1, m_l = 0$$

(2014)

©

Options:

A. 1

B. 2

C. 3

D. 4

Answer: A

Solution:

Solution:

Only one orbital, $3p_z$ has following set of quantum numbers, $n = 3$, $l = 1$ and $m_l = 0$

Question31

The values of Planck's constant is 6.63×10^{-34} J s. The speed of light is 3×10^{17} nms⁻¹. Which value is closet to the wave length in nanometer of quantum of light with frequency of 6×10^{15} s⁻¹ (2013 NEET)

©

Options:

- A. 50
- B. 75
- C. 10
- D. 25

Answer: A

Solution:

Solution:

$$c = \nu \lambda \Rightarrow \lambda = \frac{c}{\nu} = \frac{3 \times 10^{17}}{6 \times 10^{15}} = 50 \text{ nm}$$

Question32

Based on equation $E = -2.178 \times 10^{-18} \text{ J} \left(\frac{Z^2}{n^2} \right)$, certain conclusion are written. Which of them is not correct ? (2013 NEET)

©

Options:

- A. Equation can be used to calculate the change in energy when the electron changes orbit
- B. For $n = 1$, the electron has a more negative energy than it does for $n = 6$ which means that the electron is more loosely bound in the smallest allowed orbit
- C. The negative sign in equation simply means that the energy of electron bound to the nucleus is lower than it would be if the electrons were at the infinite distance from nucleus
- D. Larger the value of n , the larger is the orbit radius

Answer: B

Solution:

Solution:

The electron is more tightly bound in the smallest allowed orbit.

Question33

What is the maximum numbers of electrons that can be associated with the following set of quantum numbers ? $n = 3$, $l = 1$ and $m = -1$ (2013 NEET)

©

Options:

- A. 4
- B. 2
- C. 10
- D. 6

Answer: B

Solution:

Solution:

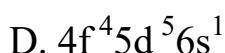
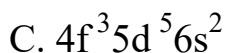
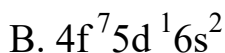
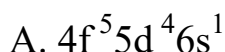
The orbital associated with $n = 3, l = 1$ is 3p. One orbital (with $m = -1$) of 3p subshell can accommodate maximum 2 electrons.

Question34

The outer electronic configuration of Gd (At. No. 64) is (Karnataka NEET 2013)

©

Options:



Answer: B

Solution:

Solution:

The electronic configuration of ${}_{64}\text{Gd}$ is $[\text{Xe}]4f^7 5d^1 6s^2$

Question35

According to law of photochemical equivalence the energy absorbed (in ergs/mole) is given as ($h = 6.62 \times 10^{-27}$ ergs, $c = 3 \times 10^{10} \text{ cms}^{-1}$, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$) (Karnataka NEET 2013)

©

Options:

A. $\frac{1.196 \times 10^8}{\lambda}$

B. $\frac{2.859 \times 10^5}{\lambda}$

C. $\frac{2.859 \times 10^{16}}{\lambda}$

D. $\frac{1.196 \times 10^{16}}{\lambda}$

Answer: A

Solution:

Solution:

$$\begin{aligned} \text{We know that, } E &= \frac{hcN_A}{\lambda} \\ &= \frac{6.62 \times 10^{-27} \times 3 \times 10^{10} \times 6.02 \times 10^{23}}{\lambda} \\ &= \frac{1.1955 \times 10^8}{\lambda} = \frac{1.196 \times 10^8}{\lambda} \text{ ergs mol}^{-1} \end{aligned}$$

Question36

The orbital angular momentum of p-electron is given as (2012 Mains)

©

Options:

A. $\frac{h}{\sqrt{2}\pi}$

B. $\sqrt{3} \frac{h}{2\pi}$

C. $\sqrt{\frac{3}{2}} \frac{h}{\pi}$

D. $\sqrt{6} \frac{h}{2\pi}$

Answer: A

Solution:

Solution:

$$\text{Orbital angular momentum}(m) = \sqrt{l(l+1)} \frac{h}{2\pi}$$

For p-electrons, $l = 1$

Thus,

$$m = \sqrt{1(1+1)} \frac{h}{2\pi} = \frac{\sqrt{2}h}{2\pi} = \frac{h}{\sqrt{2}\pi}$$

Question37

The correct set of four quantum number for the valence electron of rubidium atom ($Z = 37$) is (2012)

©

Options:

A. $5, 1, 1, +\frac{1}{2}$

B. $6, 0, 0, +\frac{1}{2}$

C. $5, 0, 0, +\frac{1}{2}$

D. $5, 1, 0, +\frac{1}{2}$

Answer: C

Solution:

Solution:

$$\text{Rb}(37) : 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1$$

$$\text{For } 5s, n = 5, l = 0, m = 0, s = +\frac{1}{2} \text{ or } -\frac{1}{2}$$

Question38

Maximum number of electrons in a subshell with $l = 3$ and $n = 4$ is (2012)

Options:

- A. 14
- B. 16
- C. 10
- D. 12

Answer: A**Solution:**

Solution:

$l = 3$ and $n = 4$ represent 4f. So total number of electrons in a subshell $= 2(2l + 1) = 2(2 \times 3 + 1) = 14$ electrons, Hence f-subshell can contain maximum 14 electrons.

Question39

**According to the Bohr theory, which of the following transition in the hydrogen atom will give rise to the least energetic photon ?
(2011 Mains)**

Options:

- A. $n = 6$ to $n = 1$
- B. $n = 5$ to $n = 4$
- C. $n = 6$ to $n = 5$
- D. $n = 5$ to $n = 3$

Answer: C**Solution:**

Solution:

We know that

$$\Delta E \propto \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right], \text{ where } n_2 > n_1$$

$\therefore n = 6$ to $n = 5$ will give least energetic photon

Question40

If $n = 6$, the correct sequence for filling of electrons will be (2011)

©

Options:

A.

$$ns \rightarrow (n-2)f \rightarrow (n-1)d \rightarrow np$$

B.

$$ns \rightarrow (n-1)d \rightarrow (n-2)f \rightarrow np$$

C.

$$ns \rightarrow (n-2)f \rightarrow np \rightarrow (n-1)d$$

D.

$$ns \rightarrow np(n-1)d \rightarrow (n-2)f$$

Answer: A

Solution:

Solution:

Question41

The energies E_1 and E_2 of two radiations are 25 eV and 50 eV respectively. The relations between their wavelengths i.e., λ_1 and λ_2 will be (2011)

Options:

A. $\lambda_1 = \lambda_2$

B. $\lambda_1 = 2\lambda_2$

C. $\lambda_1 = 4\lambda_2$

D. $\lambda_1 = \frac{1}{2}\lambda_2$

Answer: B

Solution:

Solution:

$$E_1 = \frac{hc}{\lambda_1} \text{ and } E_2 = \frac{hc}{\lambda_2};$$

$$\frac{E_1}{E_2} = \frac{hc}{\lambda_1} \times \frac{\lambda_2}{hc} = \frac{\lambda_2}{\lambda_1}$$

$$\text{or } \frac{25}{50} = \frac{\lambda_2}{\lambda_1} \text{ or } \frac{1}{2} = \frac{\lambda_2}{\lambda_1} \Rightarrow \lambda_1 = 2\lambda_2$$

Question42

**The total number of atomic orbitals in fourth energy level of an atom is
(2011)**

Options:

A. 8

B. 16

C. 32

D. 4

Answer: B

Solution:

Solution:

Total number of atomic orbitals in a any energy level is given by n^2

Question43

A 0.66 kg ball is moving with a speed of 100 m/s. The associated wavelength will be ($h = 6.6 \times 10^{-34} \text{ J s}$) (2010 Mains)

©

Options:

A. $6.6 \times 10^{-32} \text{ m}$

B. $6.6 \times 10^{-34} \text{ m}$

C. $1.0 \times 10^{-35} \text{ m}$

D. $1.0 \times 10^{-32} \text{ m}$

Answer: C

Solution:

Solution:

According to de-Broglie equation, $\lambda = \frac{h}{mv}$

Given $h = 6.6 \times 10^{-34} \text{ J s}$

$m = 0.66 \text{ kg}$ $v = 100 \text{ ms}^{-1}$

$$\therefore \lambda = \frac{6.6 \times 10^{-34}}{0.66 \times 100} = 1 \times 10^{-35} \text{ m}$$

Question44

Which of the following is not permissible arrangement of electrons in an atom ?

(2009)

©

Options:

A.

$$n = 5, l = 3, m = 0, s = +\frac{1}{2}$$

B.

$$n = 3, l = 2, m = -3, s = -\frac{1}{2}$$

C.

$$n = 3, l = 2, m = -2, s = -\frac{1}{2}$$

D.

$$n = 4, l = 0, m = 0, s = -\frac{1}{2}$$

Answer: B

Solution:

Solution:

In an atom, for any value of n , the value of l is 0 to $(n - 1)$.

For a given value of l the values of $m_l = -l$ to 0 to $+l$ and the value of $s = +\frac{1}{2}$ or $-\frac{1}{2}$

In option (b), $l = 2$ and $m_l = -3$

This is not possible as values of m_l which are possible for $l = 2$ are $-2, -1, 0, +1$ and $+2$ only

Question 45

Maximum number of electrons in a sub shell of an atom is determined by the following

(2009)

©

Options:

A. $2l + 1$

B. $4l - 2$

C. $2n^2$

D. $4l + 2$

Answer: D

Solution:

Solution:

For a given shell, l ,

the number of sub shells, $m_l = (2l + 1)$

Since each subshell can accommodate 2 electrons of opposite spin, so maximum number of electrons in a subshell

$$= 2(2l + 1) = 4l + 2$$

Question 46

The measurement of the electron position is associated with an uncertainty in momentum, which is equal to $1 \times 10^{-18} \text{ g cm s}^{-1}$. The uncertainty in electron velocity is (mass of an electron is $9 \times 10^{-28} \text{ g}$) (2008)

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Options:

A. $1 \times 10^5 \text{ cm s}^{-1}$

B. $1 \times 10^{11} \text{ cm s}^{-1}$

C. $1 \times 10^9 \text{ cm s}^{-1}$

D. $1 \times 10^6 \text{ cm s}^{-1}$

Answer: C

Solution:

Solution:

Uncertainty in momentum ($m \Delta v$) = $1 \times 10^{-18} \text{ g cm s}^{-1}$

Uncertainty in velocity (Δv) = $\frac{1 \times 10^{-18}}{9 \times 10^{-28}} = 1.1 \times 10^9 \text{ cm s}^{-1}$

Question47

If uncertainty in position and momentum are equal ,then uncertainty in velocity is (2008)

©

Options:

A. $\frac{1}{m} \sqrt{\frac{\hbar}{\pi}}$

B. $\sqrt{\frac{\hbar}{\pi}}$

C. $\frac{1}{2m} \sqrt{\frac{\hbar}{\pi}}$

D. $\sqrt{\frac{\hbar}{2\pi}}$

Answer: C

Solution:

Solution:

From Heisenberg's uncertainty principle

$$\Delta p \cdot \Delta x \geq \frac{\hbar}{4\pi} \text{ or } m\Delta v \times \Delta x \geq \frac{\hbar}{4\pi}$$

$$\text{or } (m\Delta v)^2 \geq \frac{\hbar}{4\pi} \quad (\because \Delta x = \Delta p)$$

$$\text{or } \Delta v \geq \frac{1}{2m} \sqrt{\frac{\hbar}{\pi}}$$

Question48

Consider the following sets of quantum numbers :

n l m s

(i) 3 0 0 +1/2

(ii) 2 2 1 +1/2

(iii) 4 3 -2 -1/2

(iv) 1 0 -1 -1/2

(v) 3 2 3 +1/2

**Which of the following sets of quantum number is not possible ?
(2007)**

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Options:

A. (i), (ii), (iii) and (iv)

B. (ii), (iv) and (v)

C. (i), and (iii)

D. (ii), (iii) and (iv)

Answer: B

Solution:

Solution:

(i) represents an electron in 3s orbital

(ii) is not possible as value of l varies from 0, 1, ... (n - 1).

(iii) represents an electron in 4f orbital

(iv) is not possible as value of m varies from -1 +1

(v) is not possible as value of m varies from -1 +1, it can never be greater than l

Question 49

**Given : The mass of electron is 9.11×10^{-31} kg, Planck's constant is 6.626×10^{-34} Js, the uncertainty involved in the measurements of velocity with in a distance of 0.1 Å is
(2006)**

©

Options:

A. $5.79 \times 10^5 \text{ms}^{-1}$

B. $5.79 \times 10^6 \text{ms}^{-1}$

C. $5.79 \times 10^7 \text{ms}^{-1}$

D. $5.79 \times 10^8 \text{ms}^{-1}$

Answer: B

Solution:

Solution:

$$\Delta x \cdot m\Delta v = \frac{h}{4\pi}$$

$$0.1 \times 10^{-10} \times 9.11 \times 10^{-31} \times \Delta v = \frac{6.626 \times 10^{-34}}{4 \times 3.143}$$

$$\therefore \Delta v = \frac{6.626 \times 10^{-34}}{0.1 \times 10^{-10} \times 9.11 \times 10^{-31} \times 4 \times 3.143} = 5.79 \times 10^6 \text{ms}^{-1}$$

Question50

The orientation of an atomic orbital is governed by (2006)

©

Options:

A. pricipal quantum number

B. azimuthal quantum number

C. spin quantum number

D. magnetic quantum number

Answer: D

Solution:

Solution:

Principal quantum number represents the name,size and energy of the shell to which the electron belongs.

Azimuthal quantum number describes the spatial distribution of electron cloud and angular

momentum.

Magnetic quantum number describes the orientation or distribution of electron

Spin quantum number represents the direction of electron spin around its own axis

Question51

The energy of second Bohr orbit of the hydrogen atom is -328kJ mol^{-1} ; hence the energy of fourth Bohr orbit would be (2005)

©

Options:

A. -41kJ mol^{-1}

B. -82kJ mol^{-1}

C. -164kJ mol^{-1}

D. -1312kJ mol^{-1}

Answer: B

Solution:

Solution:

$$E_n = -K \left(\frac{Z}{n} \right)^2$$

$Z = 1$ for hydrogen; $n = 2$

$$E_2 = \frac{-K \times 1}{4} \Rightarrow E_2 = -328\text{kJ mol}^{-1}; K = 4 \times 328$$

$$E_4 = \frac{-K \times 1}{16} \Rightarrow E_4 = -4 \times 328 \times \frac{1}{16} = -82\text{kJ mol}^{-1}$$

Question52

The frequency of radiation emitted when the electron falls from $n = 4$ to $n = 1$ in a hydrogen atom will be (Given ionization energy of $\text{H} = 2.18 \times 10^{-18}\text{J atom}^{-1}$ and $h = 6.625 \times 10^{-34}\text{J s}$) (2004)

Options:

A. $1.54 \times 10^{15} \text{ s}^{-1}$

B. $1.03 \times 10^{15} \text{ s}^{-1}$

C. $3.08 \times 10^{15} \text{ s}^{-1}$

D. $2.00 \times 10^{15} \text{ s}^{-1}$

Answer: C

Solution:

Solution:

$$E = h\nu \text{ or } \nu = E/h$$

$$\text{For H atom, } E = \frac{-21.76 \times 10^{-19}}{n^2} \text{ J atm}^{-1}$$

$$\Delta E = -21.76 \times 10^{-19} \left(\frac{1}{4^2} - \frac{1}{1^2} \right)$$

$$= 20.40 \times 10^{-19} \text{ J atm}^{-1}$$

$$\nu = \frac{20.40 \times 10^{-19}}{6.626 \times 10^{-34}} = 3.079 \times 10^{15} \text{ s}^{-1}$$

Question 53

The value of Planck's constant is $6.63 \times 10^{-34} \text{ J s}$. The velocity of light is $3.0 \times 10^8 \text{ ms}^{-1}$. Which value is closest to the wavelength in nanometers of a quantum of light with frequency of $8 \times 10^{15} \text{ s}^{-1}$? (2003)

Options:

A. 2×10^{-25}

B. 5×10^{-18}

C. 4×10^1

D. 3×10^7

Answer: C

Solution:

Solution:

Applying $v = c/\lambda$

$$\lambda = \frac{c}{v} = \frac{3 \times 10^8}{8 \times 10^{15}} = 37.5 \times 10^{-9} \text{m}$$

$$= 37.5 \text{nm} \approx 4 \times 10^1 \text{nm}$$

Question54

In hydrogen atom, energy of first excited state is -3.4eV . Then find out K . E . of same orbit of hydrogen atom (2002)

©

Options:

A. $+3.4\text{eV}$

B. $+6.8\text{eV}$

C. -13.6eV

D. $+13.6\text{eV}$

Answer: A

Solution:

Solution:

$$\text{Kinetic energy} = \frac{1}{2}mv^2 = \left(\frac{\pi e^2}{nh} \right)^2 \times 2m$$

$$\left[\because v = \frac{2\pi e^2}{nh} \right]$$

Total energy

$$E_n = -\frac{2\pi^2 m e^4}{n^2 h^2} = -\left(\frac{\pi e^2}{nh}\right)^2 \times 2m = -K \cdot E$$

$\therefore$ Kinetic energy $= -E_n$

Energy of first excited state is -3.4eV .

$\therefore$ Kinetic energy of same orbit ($n = 2$) will be $+3.4\text{eV}$.

Question55

Main axis of a diatomic molecule is z, molecular orbital p_x and p_y overlap to form which of the following orbitals. (2001)

Options:

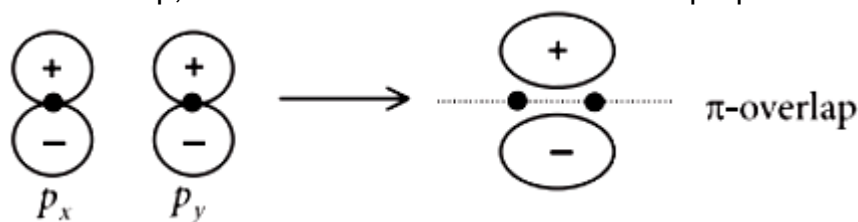
- A. π molecular orbital
- B. σ molecular orbital
- C. δ molecular orbital
- D. No bond will form.

Answer: A

Solution:

Solution:

For π overlap, the lobes of the atomic orbitals are perpendicular to the line joining the nuclei.



Hence, only sidewise overlapping takes place.

Question56

The following quantum numbers are possible for how many orbitals? $n = 3, l = 2, m = +2$ (2001)

Options:

- A. 1
- B. 2
- C. 3
- D. 4

Answer: A

Solution:

Solution:

$$n = 3, l = 2, m = +2$$

It symbolises one of the five d -orbitals (3d).

$m = +2$	$+1$	0	-1	-2

Question57

For given energy, $E = 3.03 \times 10^{-19}$ joules corresponding wavelength is

($h = 6.626 \times 10^{-34}$ J sec, $c = 3 \times 10^8$ m/sec)
(2000)

©

Options:

- A. 65.6nm
- B. 6.56nm
- C. 3.4nm
- D. 656nm

Answer: D

Solution:

Solution:

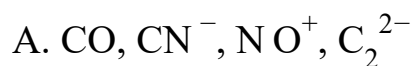
$$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{3.03 \times 10^{-19}}$$
$$= 656\text{nm}$$

Question58

**Isoelectronic species are
(2000)**

©

Options:



Answer: A

Solution:

Solution:

Species having same no. of electrons are called isoelectronics.

The no. of electrons in $\text{CO} = \text{CN}^- = \text{NO}^+ = \text{C}_2^{2-} = 14$

So, these are isoelectronic species.

Question59

**The uncertainty in momentum of an electron is $1 \times 10^{-5} \text{kgm/s}$. The uncertainty in its position will be ($h = 6.62 \times 10^{-34} \text{kgm}^2/\text{s}$)
(1999)**

©

Options:

A. $5.27 \times 10^{-30} \text{ m}$

B. $1.05 \times 10^{-26} \text{ m}$

C. $1.05 \times 10^{-28} \text{ m}$

D. $5.25 \times 10^{-28} \text{ m}$

Answer: A

Solution:

Solution:

$$\Delta x \times \Delta p = \frac{h}{4\pi}$$

(Heisenberg uncertainty principle)

$$\Rightarrow \Delta x = \frac{6.62 \times 10^{-34}}{4 \times 3.14 \times 10^{-5}} = 5.27 \times 10^{-30} \text{ m}$$

Question60

**Who modified Bohr's theory by introducing elliptical orbits for electron path?
(1999)**

Options:

A. Rutherford

B. Thomson

C. Hund

D. Sommerfeld

Answer: D

Solution:

Solution:

Sommerfeld modified Bohr's theory considering that in addition to circular orbits electrons also move in elliptical orbits.

Question61

The de Broglie wavelength of a particle with mass 1g and velocity 100m/s is
(1999)

©

Options:

A. $6.63 \times 10^{-35} \text{ m}$

B. $6.63 \times 10^{-34} \text{ m}$

C. $6.63 \times 10^{-33} \text{ m}$

D. $6.65 \times 10^{-35} \text{ m}$

Answer: C

Solution:

Solution:

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-27} \text{ erg sec}}{1 \text{ g} \times 10^4 \text{ cm/s}} \\ = 6.63 \times 10^{-31} \text{ cm} = 6.63 \times 10^{-33} \text{ m}$$

Question62

The Bohr orbit radius for the hydrogen atom ($n = 1$) is approximately $0.530 \text{ \AA}$. The radius for the first excited state ($n = 2$) orbit is (in $\text{\AA}$)
(1998)

©

Options:

A. 4.77

B. 1.06

C. 0.13

D. 2.12

Answer: D

Solution:

Solution:

For n^{th} orbit of 'H' atom, $r_n = n^2 \times r_1$

$\Rightarrow$ radius of 2^{nd} Bohr's orbit.

$$r_2 = 4 \times r_1 = 4 \times 0.530 = 2.120 \text{ \AA}$$

Question63

The position of both, an electron and a helium atom is known within 1.0nm Further the momentum of the electron is known within $5.0 \times 10^{-26} \text{ kgms}^{-1}$. The minimum uncertainty in the measurement of the momentum of the helium atom is (1998)

©

Options:

A. $8.0 \times 10^{-26} \text{ kgms}^{-1}$

B. 80 kgms^{-1}

C. 50 kgms^{-1}

D. $5.0 \times 10^{-26} \text{ kgms}^{-1}$

Answer: D

Solution:

Solution:

According to uncertainty principle the product of uncertainty in position and uncertainty in momentum is constant for a particle.

i.e., $\Delta x \times \Delta p = \frac{h}{4\pi}$

As, $\Delta x = 1.0\text{nm}$ for both electron and helium atom, so Δp is also same for both the particles. Thus uncertainty in momentum of the helium atom is also $5.0 \times 10^{-26}\text{kgms}^{-1}$

Question64

**The ion that is isoelectronic with CO is
(1997)**

©

Options:



Answer: A

Solution:

Solution:

since both CO and CN^- have 14 electrons, therefore these are isoelectronic (i.e. having same number of electrons).

Question65

**What will be the longest wavelength line in Balmer series of
spectrum?
(1996)**

©

Options:



B. 656nm

C. 566nm

D. 556nm

Answer: B

Solution:

Solution:

The longest wavelength means the lowest energy. We know that relation for wavelength

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

here, $n_1 = 2$, $n_2 = 3$

R_H (Rydberg constant = 109677cm^{-1}

$$\frac{1}{\lambda} = 109677 \left(\frac{1}{(2)^2} - \frac{1}{(3)^2} \right) = 15233$$

$$\text{or, } \lambda = \frac{1}{15233} = 6.56 \times 10^{-5}\text{cm}$$

$$= 6.56 \times 10^{-7}\text{m} = 656\text{nm}$$

Question66

**In a Bohr's model of an atom, when an electron jumps from $n = 1$ to $n = 3$, how much energy will be emitted or absorbed?
(1996)**

©

Options:

A. 2.389×10^{-12} ergs

B. 0.239×10^{-10} ergs

C. 2.15×10^{-11} ergs

D. 0.1936×10^{-10} ergs

Answer: D

Solution:

Solution:

Energy of an atom when $n = 1$

$$E_1 = -\frac{1312}{(1)^2} = -1312 \text{ kJ mol}^{-1}$$

Similarly energy when $n = 3$, $(E_3) = -\frac{1312}{(3)^2}$

$$= -145.7 \text{ kJ mol}^{-1}$$

The energy absorbed when an electron jumps from $n = 1$ to $n = 3$

$$E_3 - E_1 = -145.7 - (-1312) = 1166.3 \text{ kJ mol}^{-1}$$

$$= \frac{1166.3}{6.023 \times 10^{23}} = 193.6 \times 10^{-23} \text{ kJ}$$

$$= 193.6 \times 10^{-20} \text{ J [1 joule = } 10^7 \text{ ergs]}$$

$$\Rightarrow 193.6 \times 10^{-13} \text{ ergs} = 0.1936 \times 10^{-10} \text{ ergs}$$

Question 67

Uncertainty in position of an electron (mass = $9.1 \times 10^{-28} \text{ g}$) moving with a velocity of $3 \times 10^4 \text{ cm/s}$ accurate upto 0.001% will be (Use $h/(4\pi)$ in uncertainty expression where $h = 6.626 \times 10^{-27} \text{ erg second}$) (1995)

©

Options:

A. 5.76cm

B. 7.68cm

C. 1.93cm

D. 3.84cm

Answer: C

Solution:

Solution:

Mass of an electron (m) = $9.1 \times 10^{-28} \text{ g}$

Velocity of electron (v) = $3 \times 10^4 \text{ cm/s}$

Accuracy = 0.001% = $\frac{0.001}{100}$ and Planck's constant

(h) = $6.626 \times 10^{-27} \text{ erg-second}$.

We know that actual velocity of the electron

$$(\Delta v) = 3 \times 10^4 \times \frac{0.001}{100} = 0.3 \text{ cm/s}$$

Therefore, uncertainty in the position of the electron,

$$(\Delta x) = \frac{h}{4\pi m \Delta v} = \frac{6.626 \times 10^{-27}}{4\pi \times (9.1 \times 10^{-28}) \times 0.3}$$

$$= 1.93 \text{ cm}$$

Question 68

The radius of hydrogen atom in the ground state is 0.53 Å. The radius of Li^{2+} ion (atomic number = 3) in a similar state is (1995)

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Options:

A. 0.53 Å

B. 1.06 Å

C. 0.17 Å

D. 0.265 Å

Answer: C

Solution:

Solution:

Due to ground state, state of hydrogen atom (n) = 1;

Radius of hydrogen atom (r) = 0.53 Å

Atomic no. of Li (Z) = 3

$$\text{Now, radius of } \text{Li}^{2+} \text{ ion} = r \times \frac{n^2}{Z} = 0.53 \times \frac{(1)^2}{3}$$

$$= 0.17 \text{ Å}$$

Question 69

For which of the following sets of four quantum numbers, an electron will have the highest energy?

	n	l	m	s
(a)	3	2	1	+1/2
(b)	4	2	-1	+1/2
(c)	4	1	0	-1/2
(d)	5	0	0	-1/2

(1994)

Options:

- A. a
- B. b
- C. c
- D. d

Answer: B

Solution:

Solution:

Energy of electron depends on the value of $(n + l)$ The subshell are 3d, 4d, 4p and 5s, 4d has highest energy.

Question 70

Which one of the following is not isoelectronic with O^{2-} ?
(1994)

Options:

- A. Tl^+
- B. Na^+



Answer: A

Solution:

Solution:

The number of electrons in O^{2-} , N^{3-} , F^{-} and Na^{+} is 10 each, but number of electrons in Ti^{+} is 80

Question 71

Electronic configuration of calcium atom can be written as (1992)



Options:



Answer: B

Solution:

Solution:

Atomic no. of Ca = 20

$\therefore$ Electronic configuration of Ca = $[\text{Ar}]4\text{s}^2$

Question 72

The energy of an electron in the n^{th} Bohr orbit of hydrogen atom is (1992)

Options:

A. $\frac{13.6}{n^4} \text{eV}$

B. $\frac{13.6}{n^3} \text{eV}$

C. $\frac{13.6}{n^2} \text{eV}$

D. $\frac{13.6}{n} \text{eV}$

Answer: C

Solution:

Solution:

Energy of an electron in n^{th} Bohr orbit of hydrogen atom = $\frac{-13.6}{n^2} \text{eV}$

Question 73

In a given atom no two electrons can have the same values for all the four quantum numbers. This is called (1991)

Options:

A. Hund's Rule

B. Aufbau principle

C. Uncertainty principle

D. Pauli's Exclusion principle.

Answer: D

Solution:

Solution:
This is a Pauli's exclusion principle.

Question 74

For azimuthal quantum number $l = 3$, the maximum number of electrons will be
(1991)

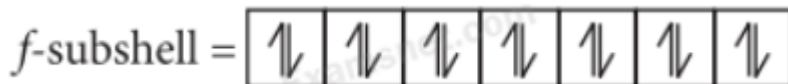
Options:

- A. 2
- B. 6
- C. 0
- D. 14

Answer: D

Solution:

Solution:
 $l = 3$ means f -subshell
Maximum no. of electrons in f -subshell = 14



Question 75

The order of filling of electrons in the orbitals of an atom will be
(1991)

Options:

- A. 3d , 4s, 4p, 4d , 5s
- B. 4s, 3d , 4p, 5s, 4d

C. 5s, 4p, 3d, 4d, 5s

D. 3d, 4p, 4s, 4d, 5s

Answer: B

Solution:

Solution:

As per Aufbau principle, in the ground state of the atoms, the orbitals are filled in order of their increasing energies.

Question 76

The electronic configuration of Cu (atomic number 29) is (1991)

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Options:

A. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9$

B. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$

C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^6 5s^2 5p^1$

D. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^6 3d^3$

Answer: B

Solution:

Solution:

Electronic configuration of Cu is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$

Question 77

The total number of electrons that can be accommodated in all the orbitals having principal quantum number 2 and azimuthal quantum number 1 are

(1990)

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Options:

A. 2

B. 4

C. 6

D. 8

Answer: C

Solution:

Solution:

$n = 2, l = 1$ It means 2p -orbitals. Total no. of electrons that can be accommodated in all the 2p orbitals = 6

Question 78

**An ion has 18 electrons in the outermost shell, it is
(1990)**

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Options:

A. Cu^+

B. Th^{4+}

C. Cs^+

D. K^+

Answer: A

Solution:

Solution:

Cu^+ ion has 18 electrons in its outermost shell. Electronic configuration of Cu^+ is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$

Question79

Which of the following statements do not form a part of Bohr's model of hydrogen atom? 10 (1989)

Options:

- A. Energy of the electrons in the orbits are quantized.
- B. The electron in the orbit nearest the nucleus has the lowest energy.
- C. Electrons revolve in different orbits around the nucleus.
- D. The position and velocity of the electrons in the orbit cannot be determined simultaneously.

Answer: D

Solution:

Solution:
It is uncertainty principle and not Bohr's postulate.

Question80

Number of unpaired electrons in N^{2+} is/are (1989)

Options:

- A. 2
- B. 0

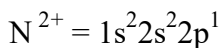
C. 1

D. 3

Answer: C

Solution:

Solution:



$\therefore$ No. of unpaired electrons = 1

Question81

The maximum number of electrons in a subshell is given by the expression
(1989)

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Options:

A. $4l - 2$

B. $4l + 2$

C. $2l + 2$

D. $2n^2$

Answer: B

Solution:

Solution:

No. of orbitals in a subshell = $2l + 1$

$\Rightarrow$ No. of electrons = $2(2l + 1) = 4l + 2$

Question82

The spectrum of He is expected to be similar to that

(1988)

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Options:

A. H

B. Li^+

C. Na

D. He^+

Answer: B

Solution:

Solution:

Both He and Li^+ contain 2 electrons each.

Question83

**The number of spherical nodes in 3p orbitals are/is
(1988)**

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Options:

A. one

B. three

C. none

D. two

Answer: A

Solution:

Solution:

No. of radial nodes in 3p -orbital $= n - l - 1$
 $= 3 - 1 - 1 = 1$

Question 84

If r is the radius of the first orbit, the radius of n^{th} orbit of H -atom is given by
(1988)

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Options:

A. $r_0 n^2$

B. $r_0 n$

C. r_0/n

D. $r_0^2 n^2$

Answer: A

Solution:

Solution:

Radius of n^{th} orbit of H-atom $= r_0 n^2$

where r_0 = radius of the first orbit.
